Bounding the PI part under the PID update.

Under PI rule ($\boldsymbol{K}_d = 0$),

$$\tilde{\zeta}_{\mathrm{PI}}(k+1)\big|_{\mathrm{PI}} = \boldsymbol{M}_i(k)\tilde{\zeta}_{\mathrm{PI}}(k) + \tilde{\boldsymbol{w}}_{\mathrm{PI}}(k), \qquad \boldsymbol{M}_i(k) := \begin{bmatrix} \boldsymbol{M}_p(k) & -\boldsymbol{G}(k) \\ I & I \end{bmatrix}$$

then

$$\begin{aligned}\Delta \boldsymbol{V}_{\mathrm{PI}}\Big(\tilde{\zeta}_{\mathrm{PI}}(k)\Big)\Big|_{\mathrm{PI}} &= \tilde{\zeta}_{\mathrm{PI}}(k)^\top \big(\boldsymbol{M}_i(k)^\top \boldsymbol{P}(k+1)\boldsymbol{M}_i(k) - \boldsymbol{P}(k)\big)\tilde{\zeta}_{\mathrm{PI}}(k) \\ &\quad + 2\tilde{\zeta}_{\mathrm{PI}}(k)^\top \boldsymbol{M}_i(k)^\top \boldsymbol{P}(k+1)\tilde{\boldsymbol{w}}_{\mathrm{PI}}(k) + \tilde{\boldsymbol{w}}_{\mathrm{PI}}(k)^\top \boldsymbol{P}(k+1)\tilde{\boldsymbol{w}}_{\mathrm{PI}}(k)\end{aligned}$$

By Eqn. 64, Cauchy-Schwarz inequality and Young's inequality,

$$\begin{aligned}\Delta \boldsymbol{V}_{\mathrm{PI}}\Big(\tilde{\zeta}_{\mathrm{PI}}(k)\Big)\Big|_{\mathrm{PI}} &\le -\mu_{\mathrm{PI}}\|\tilde{\zeta}_{\mathrm{PI}}(k)\|^2 + 2\|\boldsymbol{M}_i\|_\infty \|\boldsymbol{P}\|_\infty \|\tilde{\zeta}_{\mathrm{PI}}(k)\|\|\tilde{\boldsymbol{w}}_{\mathrm{PI}}(k)\| + \|\boldsymbol{P}\|_\infty \|\tilde{\boldsymbol{w}}_{\mathrm{PI}}(k)\|^2 \\ &\le -(\mu_{\mathrm{PI}} - \varepsilon_1)\|\tilde{\zeta}_{\mathrm{PI}}(k)\|^2 + \Big(\frac{\|\boldsymbol{M}_i\|_\infty^2 \|\boldsymbol{P}\|_\infty^2}{\varepsilon_1} + \|\boldsymbol{P}\|_\infty\Big)\|\tilde{\boldsymbol{w}}_{\mathrm{PI}}(k)\|^2 \\ &= -\mu_{\mathrm{PI}}^*\|\tilde{\zeta}_{\mathrm{PI}}(k)\|^2 + C_1\|\tilde{\boldsymbol{w}}_{\mathrm{PI}}(k)\|^2,\end{aligned}$$

for any $\varepsilon_1 > 0$.

Under PID rule ($\boldsymbol{K}_d \neq 0$),

$$\tilde{\zeta}_{\mathrm{PI}}(k+1)\big|_{\mathrm{PID}} = \boldsymbol{M}_i(k)\tilde{\zeta}_{\mathrm{PI}}(k) - \delta(k) + \boldsymbol{w}_{\mathrm{PI}}(k),$$

with the "perturbation"

$$\delta(k) := \begin{bmatrix} H(k)\Delta\bar{\boldsymbol{e}}(k) \\ 0 \end{bmatrix}$$

Hence,

$$\begin{aligned}\Delta \boldsymbol{V}_{\mathrm{PI}}\big(\tilde{\zeta}_{\mathrm{PI}}(k)\big)\Big|_{\mathrm{PID}} &= \boldsymbol{V}_{\mathrm{PI}}\big(\tilde{\zeta}_{\mathrm{PI}}(k+1)\big|_{\mathrm{PI}}\big) - \boldsymbol{V}_{\mathrm{PI}}\big(\tilde{\zeta}_{\mathrm{PI}}(k)\big) \\ &= \boldsymbol{V}_{\mathrm{PI}}\big(\tilde{\zeta}_{\mathrm{PI}}(k+1)\big|_{\mathrm{PI}}\big) - \boldsymbol{V}_{\mathrm{PI}}\big(\tilde{\zeta}_{\mathrm{PI}}(k)\big) + 2\big(\tilde{\zeta}_{\mathrm{PI}}(k+1)\big|_{\mathrm{PI}}\big)^\top \boldsymbol{P}\delta(k) + \delta(k)^\top \boldsymbol{P}\delta(k)\end{aligned} \tag{68}$$

Bounding each term in Eqn. 68

- $\boldsymbol{V}_{\mathrm{PI}}\big(\tilde{\zeta}_{\mathrm{PI}}(k+1)\big|_{\mathrm{PI}}\big) - \boldsymbol{V}_{\mathrm{PI}}\big(\tilde{\zeta}_{\mathrm{PI}}(k)\big) \le -\mu_{\mathrm{PI}}^*\|\tilde{\zeta}_{\mathrm{PI}}(k)\|^2 + C_1\|\tilde{\boldsymbol{w}}_{\mathrm{PI}}(k)\|$
- Applying Young's inequality for inner product, there exists $\varepsilon > 0$ s.t

$$2\big(\tilde{\zeta}_{\mathrm{PI}}(k+1)\big|_{\mathrm{PI}}\big)^\top \boldsymbol{P}\delta(k) \le \varepsilon\big\|\tilde{\zeta}_{\mathrm{PI}}(k+1)\big|_{\mathrm{PI}}\big\|^2 + \tfrac{1}{\varepsilon}\delta(k)^\top \boldsymbol{P}\delta(k) \tag{69}$$

$$\le \varepsilon\|\boldsymbol{P}\|\big\|\tilde{\zeta}_{\mathrm{PI}}(k+1)\big|_{\mathrm{PI}}\big\|^2 + \tfrac{\|\boldsymbol{P}\|}{\varepsilon}M^2\ell^2\|\Delta\bar{\boldsymbol{e}}(k)\|^2 \tag{70}$$

  Since $\big\|\tilde{\zeta}_{\mathrm{PI}}(k+1)\big|_{\mathrm{PI}}\big\|^2 \le 2\|\boldsymbol{M}_i\|_\infty^2\|\tilde{\zeta}_{\mathrm{PI}}(k)\|^2 + 2\|\tilde{\boldsymbol{w}}_{\mathrm{PI}}(k)\|^2$

$$\begin{aligned}\Rightarrow 2\big(\tilde{\zeta}_{\mathrm{PI}}(k+1)\big|_{\mathrm{PI}}\big)^\top \boldsymbol{P}\delta(k) &\le 2\varepsilon\|\boldsymbol{P}\|\|\boldsymbol{M}_i\|_\infty^2\|\tilde{\zeta}_{\mathrm{PI}}(k)\|^2 + 2\varepsilon\|\boldsymbol{P}\|\|\tilde{\boldsymbol{w}}_{\mathrm{PI}}(k)\|^2 \\ &\quad + \tfrac{\|\boldsymbol{P}\|}{\varepsilon}M^2\ell^2\|\Delta\bar{\boldsymbol{e}}(k)\|^2 \\ &= 2\varepsilon\|\boldsymbol{P}\|\|\boldsymbol{M}_i\|_\infty^2\|\tilde{\zeta}_{\mathrm{PI}}(k)\|^2 + \tfrac{\|\boldsymbol{P}\|}{\varepsilon}M^2\ell^2\|\Delta\bar{\boldsymbol{e}}(k)\|^2 \\ &\quad + C_2\|\tilde{\boldsymbol{w}}_{\mathrm{PI}}(k)\|^2,\end{aligned}$$

  where $C_2 = 2\varepsilon\|\boldsymbol{P}\|$
- $\delta(k)^\top \boldsymbol{P}\delta(k) \le \|\boldsymbol{P}\|\|\delta(k)\|^2 \le \|\boldsymbol{P}\|M^2\ell^2\|\Delta\bar{\boldsymbol{e}}(k)\|^2$

Therefore,

$$\begin{aligned}\Delta \boldsymbol{V}_{\mathrm{PI}}\big(\tilde{\zeta}_{\mathrm{PI}}(k)\big)\Big|_{\mathrm{PID}} &\le -\mu_{\mathrm{PI}}^*\|\tilde{\zeta}_{\mathrm{PI}}(k)\|^2 + C_1\|\tilde{\boldsymbol{w}}_{\mathrm{PI}}(k)\|^2 \\ &\quad + 2\varepsilon\|\boldsymbol{P}\|\|\boldsymbol{M}_i\|_\infty^2\|\tilde{\zeta}_{\mathrm{PI}}(k)\|^2 + \tfrac{\|\boldsymbol{P}\|}{\varepsilon}M^2\ell^2\|\Delta\bar{\boldsymbol{e}}(k)\|^2 + C_2\|\tilde{\boldsymbol{w}}_{\mathrm{PI}}(k)\|^2 \\ &\quad + \|\boldsymbol{P}\|M^2\ell^2\|\Delta\bar{\boldsymbol{e}}(k)\|^2 \\ &= -(\mu_{\mathrm{PI}}^* - 2\varepsilon\|\boldsymbol{P}\|\|\boldsymbol{M}_i\|_\infty^2)\|\tilde{\zeta}_{\mathrm{PI}}(k)\|^2 + \|\boldsymbol{P}\|M^2\ell^2\Big(\tfrac{1}{\varepsilon}+1\Big)\|\Delta\bar{\boldsymbol{e}}(k)\|^2 \\ &\quad + C_3\|\tilde{\boldsymbol{w}}_{\mathrm{PI}}(k)\|^2,\end{aligned} \tag{71}$$

where $C_3 = C_1 + C_2$.

Bounding the increment term.

From Eqn. 59 and applying the inequality $(x+y+z)^2 \le 3(x^2+y^2+z^2)$,

$$\|\Delta\bar{\boldsymbol{e}}(k+1)\|^2 \le 3\Big(\|\boldsymbol{M}_p(k)-I\|_\infty^2 + Mh\Big)\|\tilde{\zeta}_{\mathrm{PI}}(k)\|^2 + 3M^2\ell^2\|\Delta\bar{\boldsymbol{e}}(k)\|^2 + 3\|\tilde{\boldsymbol{w}}_{\mathrm{PID}}(k)\|^2,$$

and so

$$\begin{aligned} r\big(\|\Delta\bar{\boldsymbol{e}}(k+1)\|^2 - \|\Delta\bar{\boldsymbol{e}}(k)\|^2\big) \le{}& 3r\Big(\|\boldsymbol{M}_p(k)-I\|_\infty^2 + Mh\Big)\|\tilde{\zeta}_{\mathrm{PI}}(k)\|^2 \\ &- r\big(1-3M^2\ell^2\big)\|\Delta\bar{\boldsymbol{e}}(k)\|^2 + 3r\|\tilde{\boldsymbol{w}}_{\mathrm{PID}}(k)\|^2. \end{aligned} \tag{72}$$

Combination

Combining Eqn. 67, Ineq. 71, and Ineq. 72,

$$\begin{aligned} \Delta\boldsymbol{V}_{\mathrm{PID}}(k) \le{}& -\Big(\mu_{\mathrm{PI}}^* - 2\varepsilon\|\boldsymbol{P}\|_\infty\|\boldsymbol{M}_i\|_\infty^2 - 3r\big(\|\boldsymbol{M}_p(k)-I\|_\infty^2 + Mh\big)\Big)\|\tilde{\zeta}_{\mathrm{PI}}(k)\|^2 \\ &- \Big(r(1-3M^2\ell^2) - \|\boldsymbol{P}\|_\infty M^2\ell^2\Big(\tfrac{1}{\varepsilon}+1\Big)\Big)\|\Delta\bar{\boldsymbol{e}}(k)\|^2 + C\|\tilde{\boldsymbol{w}}_{\mathrm{PI}}(k)\|^2, \end{aligned}$$

where $C = C_3 + r$. Define

$$S(r,\varepsilon) := \mu_{\mathrm{PI}}^* - 2\varepsilon\|\boldsymbol{P}\|_\infty\|\boldsymbol{M}_i\|_\infty^2 - 3r\big(\|\boldsymbol{M}_p(k)-I\|_\infty^2 + Mh\big), \tag{73}$$

$$T(r,\varepsilon,\ell) := r(1-3M^2\ell^2) - \|\boldsymbol{P}\|_\infty M^2\ell^2\Big(\tfrac{1}{\varepsilon}+1\Big). \tag{74}$$

ISS of the PID loop follows if $S(r,\varepsilon) > 0$ and $T(r,\varepsilon,\ell) > 0$.

Feasible choices.

We are free to choose any $\varepsilon > 0$ and $r > 0$ such that $S(r,\varepsilon) > 0$. One convenient selection is

$$\varepsilon = \frac{\mu_{\mathrm{PI}}^*}{8\|\boldsymbol{P}\|_\infty\|\boldsymbol{M}_i\|_\infty^2}, \qquad r = \frac{\mu_{\mathrm{PI}}^*}{8\big(\|\boldsymbol{M}_p(k)-I\|_\infty^2 + Mh\big)} \quad \Rightarrow \quad S(r,\varepsilon) = \tfrac{3}{8}\mu_{\mathrm{PI}}^* > 0.$$

With ε, r fixed as above, pick $\ell > 0$ small enough to satisfy $T(r,\varepsilon,\ell) > 0$, namely

$$\ell^2 < \frac{r}{\big(\|\boldsymbol{P}\|_\infty\big(\frac{1}{\varepsilon}+1\big) + 3r\big)M^2},$$

Under these choices, $\Delta\boldsymbol{V}_{\mathrm{PID}}(k) \le -\alpha_3\|\zeta_{\mathrm{PI}}(k)\|^2 - \alpha_4\|\Delta\bar{\boldsymbol{e}}(k)\|^2 + \beta\|\boldsymbol{w}_{\mathrm{PI}}(k)\|^2$ for some $\alpha_3, \alpha_4, \beta > 0$, which satisfies condition (ii) as in Eqn. 62 and proves ISS of the PID closed loop. □

B.7.2 Overshooting under PID law of control

Developing from Sec. B.6, we introduce scalar PID recursion along v:

$$\boldsymbol{e}_v(k+1) = a(k)\boldsymbol{e}_v(k) - b(k)\boldsymbol{s}_v(k) - c(k)\Delta\boldsymbol{e}_v(k) + \boldsymbol{w}_v^\perp(k), \qquad \boldsymbol{s}_v(k+1) = \boldsymbol{s}_v(k) + \boldsymbol{e}_v(k) - d_v(k), \tag{75}$$

where

$$a(k) := v^\top\boldsymbol{M}_p(k)v, \qquad b(k) := v^\top\boldsymbol{G}(k)v, \qquad c(k) := v^\top\boldsymbol{H}(k)v,$$
$$\Delta\boldsymbol{e}_v(k) = v^\top\Delta\bar{\boldsymbol{e}}(k), \quad \boldsymbol{w}_v^\perp(k) = v^\top\boldsymbol{w}^\perp(k).$$

By construction $a(k) \le q < 1$, $b(k) \le Mh$, $c(k) \le M\ell$ with $M := \sup_k\|\bar{\boldsymbol{A}}(k)\|$, $h := \|\boldsymbol{K}_i\|$ and $\ell := \|\boldsymbol{K}_d\|$.

We now impose an additional requirement on the derivative gain $\boldsymbol{K}_d$ so that, without the effect of noise, the PID update secures the monotonic decrease of $\boldsymbol{e}_v(k)$ before the first negative peak of scalar error $\boldsymbol{e}_v(k)$.

Remark 4 (Pre-overshoot monotonic decrease of scalar errors) *Assume the setting of Proposition 4 and further suppose the scalar error trajectory before the first largest overshoot under PID law is* smooth *in the sense that there exists $R \ge 1$ such that*

$$\frac{\boldsymbol{e}_v(k-1)}{\boldsymbol{e}_v(k)} \le R \qquad \textit{for all } k = 1,2,\ldots,i_{max}-1, \tag{76}$$

where $\boldsymbol{A}_0 := \max_{k_0 \le i \le k_1} |\boldsymbol{e}_v(i)| = |\boldsymbol{e}_v(i_{max})|$ *from Eqn.53.*

Assume that $\boldsymbol{w}_v^{\perp} \equiv 0$ *and* $\boldsymbol{d}_v \equiv 0$.

If, in addition, the derivative gain satisfies

$$l = \|\boldsymbol{K}_d\| \le \frac{1-q}{(R-1)M}, \tag{77}$$

then under PID law

$$\boldsymbol{e}_v(k+1) \le \boldsymbol{e}_v(k) \qquad \textit{for all } k = 0,1,\ldots,i_{max}-1.$$

Proof. Before k_0, we have $\boldsymbol{e}_v(k) > 0$, so $\boldsymbol{s}_v(k) \ge 0$ (since $\boldsymbol{s}_v$ accumulates $\boldsymbol{e}_v$ and $\boldsymbol{s}_v(0) = 0$). For $k_0 \le k \le i_{max}-1$, $\boldsymbol{s}_v(k) \ge 0$ proved in remark 5

Hence

$$\begin{aligned}\boldsymbol{e}_v(k+1) &= a(k)\boldsymbol{e}_v(k) - b(k)\boldsymbol{s}_v(k) - c(k)\big(\boldsymbol{e}_v(k) - \boldsymbol{e}_v(k-1)\big)\\ &\le a(k)\boldsymbol{e}_v(k) + c(k)\big(\boldsymbol{e}_v(k-1) - \boldsymbol{e}_v(k)\big)\\ &\le \Big[a(k) + c(k)(R-1)\Big]\boldsymbol{e}_v(k) \le \Big[q + (R-1)M\ell\Big]\boldsymbol{e}_v(k) \le \boldsymbol{e}_v(k),\end{aligned}$$

where the last inequality is exactly Eqn. 77. □

Note for R: In practice, one may estimate a conservative R from PI-law traces and use a small safety factor

Adding Disturbance: With bounded disturbances, the scalar update reads

$$\boldsymbol{e}_v(k+1) \le \big[q + (R-1)c_{\max}\big]\boldsymbol{e}_v(k) + |\boldsymbol{w}_v^{\perp}(k)|\,,$$

so the same one-step monotonicity conclusion holds whenever

$$|\boldsymbol{w}_v^{\perp}(k)| \le \Big(1 - \big[q + (R-1)c_{\max}\big]\Big)\boldsymbol{e}_v(k) \quad \text{for all pre-first-largest-overshooting steps.}$$

If this smallness condition on disturbances fails at some step, one-step monotonicity may be lost, but the ISS bounds proved earlier still guarantee geometric decay up to a disturbance-dependent radius.

Remark 5 (before the first negative peak, the integral state is positive) *Assume the setting of Remark. 4. Hence,*

$$\boldsymbol{s}_v(k) > 0 \qquad \textit{for all } k = k_0,\ldots,i_{\max}-1.$$

Proof. We argue by contradiction. Suppose there exists the first $\tau \in [k_0, i_{\max}-1]$ such that $\boldsymbol{s}_v(\tau) \le 0$. Then $\boldsymbol{s}_v(\tau-1) > 0$, and since we are on the first negative lobe, $\boldsymbol{e}_v(\tau) < 0$. Compute the one-step change of $\boldsymbol{e}_v$:

$$\begin{aligned}\boldsymbol{e}_v(\tau+1) - \boldsymbol{e}_v(\tau) &= (a(\tau)-1)\boldsymbol{e}_v(\tau) - b(\tau)\boldsymbol{s}_v(\tau)\\ &= (1-a(\tau))|\boldsymbol{e}_v(\tau)| + b(\tau)(-\boldsymbol{s}_v(\tau)) > 0,\end{aligned}$$

because $a(\tau) \le q < 1$, $\boldsymbol{e}_v(\tau) < 0$ and $\boldsymbol{s}_v(\tau) \le 0$. Hence $\boldsymbol{e}_v(\tau+1) > \boldsymbol{e}_v(\tau)$. By the same reasoning, as long as both $\boldsymbol{e}_v(k) < 0$ and $\boldsymbol{s}_v(k) \le 0$ hold, we have

$$\boldsymbol{e}_v(k+1) - \boldsymbol{e}_v(k) \ge (1-q)|\boldsymbol{e}_v(k)| + b_{\min}(-\boldsymbol{s}_v(k)) > 0,$$

where $b_{\min} := \inf_k b(k) > 0$. Meanwhile $\boldsymbol{s}_v(k+1) = \boldsymbol{s}_v(k) + \boldsymbol{e}_v(k) \le \boldsymbol{s}_v(k)$ on that interval, so $\boldsymbol{s}_v(k)$ is non-increasing; equivalenty $-\boldsymbol{s}_v(k)$ is non-decreasing. If $\boldsymbol{e}_v$ stayed negative forever, then $\sum_{k=0}^{N} \boldsymbol{e}_v(\tau+k) \to -\infty$, so $-\boldsymbol{s}_v(k)$ would grow without bound and the increments $\boldsymbol{e}_v(k+1) - \boldsymbol{e}_v(k)$ would eventually be arbitrarily large, forcing $\boldsymbol{e}_v$ to cross 0 in finite time. This contradicts the choice of $i_{\max}$ as the first negative peak. Therefore such τ cannot exist and $\boldsymbol{s}_v(k) > 0$ for all $k = k_0,\ldots,i_{\max}-1$. □

Theorem 2 (PID reduces the first-overshoot amplitude) *Let the first overshoot occur at index* k_0 *with amplitude* A_0 *(definition in Eqn. 53). Then, the first-overshoot amplitude under PID Steering,* A_0^{PID}*, satisfies* $A_0^{\mathrm{PID}} \le A_0^{\mathrm{PI}}$*, where* A_0^{PI} *denotes the corresponding amplitude under PI Steering.*

Proof. Under the PI law we have

$$A_0^{\mathrm{PI}} = -a(i_{max}-1)\boldsymbol{e}_v(i_{max}-1) + b(i_{max}-1)\boldsymbol{s}_v(i_{max}-1) - \boldsymbol{w}_v^{\perp}(i_{max}-1),$$